



SQL – AGGREGATION

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- מערכת ה DBMS כוללות בתוכה פונקציות מובנות המאפשרות לבצע פעולות חישוביות בדומה ליכולות של גליון אקסל.
- פעולות אלה מבוצעות ישירות על הנתונים כחלק מה SELECT

AGGREGATION FUNCTIONS IN SQL

Functions can have *arguments*, just like C, Java, etc.

SUM(*expression*)

SELECT SUM(length) FROM film;

AVG(*expression*)

SELECT AVG(length) FROM film;

MAX(*expression*)

SELECT MAX(length) FROM film;

MIN(*expression*)

SELECT MIN(length) FROM film;

COUNT(*expression*)

SELECT COUNT() FROM film;*

1. create table called *students_agg*:

```
USE sand_box;
```

```
CREATE TABLE student_agg  
(  
  student_id INT AUTO_INCREMENT,  
  student_name VARCHAR(10),  
  phone_type VARCHAR(10),  
  quantity INT,  
  PRIMARY KEY (student_id)  
);
```

2. Insert dome data:

```
INSERT INTO student_agg (student_name,phone_type,quantity) VALUES ('noam','iphone',1);  
INSERT INTO student_agg (student_name,phone_type,quantity) VALUES ('ron','android',1);  
INSERT INTO student_agg (student_name,phone_type,quantity) VALUES ('nikoli','samsung',1);  
INSERT INTO student_agg (student_name,phone_type,quantity) VALUES ('kori','xiomi',1);  
INSERT INTO student_agg (student_name,phone_type,quantity) VALUES ('yariv','xiomi',3);
```

דוגמה (המשך):

Now, query the data from the table:

```
SELECT * FROM student_agg;
```

	student_id	student_name	phone_type	quantity
	1	noam	iphone	1
	2	yariv	xiomi	3
▶	3	nikoli	samsung	1
	4	kori	xiomi	1
	5	yariv	xiomi	3
	NULL	NULL	NULL	NULL

דוגמה (המשך):

מספר הרשומות בטבלה:

```
SELECT COUNT(*) FROM student_agg;
```

מספר הטלפונים:

```
SELECT SUM(quantity) FROM student_agg;
```

GROUP BY ...

GROUP BY . . . is used when you want to apply a function (count, sum, avg) to **a group of rows** having a common characteristic.

example: How many cities are in each country?

```
SELECT country_id,COUNT(*) number_cities_in_country  
FROM city GROUP BY country_id;
```


דוגמה:

מספר הרשומות בטבלה לפי סוג טלפון:

```
SELECT PHONE_TYPE,COUNT(*) FROM student_agg GROUP BY  
phone_type;
```

מספר הטלפונים לפי סוג הטלפון:

```
SELECT PHONE_TYPE,SUM(quantity) FROM student_agg GROUP BY  
phone_type;
```

GROUP BY ... HAVING ...

(1) First, how to count number of films in each feature?

```
SELECT special_features,COUNT(*) FROM sakila.film GROUP  
BY special_features;
```

GROUP BY ... HAVING ...

(2) add HAVING to restrict results to `count( ) > 10`

```
SELECT special_features,COUNT(*) FROM sakila.film GROUP  
BY special_features HAVING COUNT(*) > 70;
```

EXERCISES

- Which first names are not repeated (actor)?
- Which last names appear more than once (actor)?
- Which actor has appeared in the most of films (film , film_actor, actor)?
- What is that average length of all the films (film)?
- What is the average length of films by category (film, film_category, category) ?
- List each film and the number of actors who are listed for that film. (film_actor, film)